SAFE-CHEM: Uncertainty-Aware Policy Switching for Robust Robotic Chemistry

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***Abstract*—...**

I. Introduction

The integration of robotics and AI has the potential to accelerate traditional human-driven chemistry experiments towards autonomous, efficient and more productive workflows [1]. The benefits of autonomous robotic chemists have been demonstrated globally across diverse workflows [2], [3], from photocatalysis [4], to solid-state materials chemistry [5]. These systems can execute a broad spectrum of experimental tasks by utilising general-purpose robots, but most remain restricted to rigid, non-reconfigurable workspaces where changing to novel experiments often incurs substantial downtime [6].

To overcome the limitations of rigid, pre-programmed traditional methods in chemistry lab automation, recent works have pivoted towards learning-based methods, specifically imitation learning (IL) and reinforcement learning (RL) [7]. Unlike traditional automation that relies on precise coordinate-based trajectories, learning-based approaches allow manipulators to acquire complex, dexterous skills *e.g.* powder weighing [8], [9], pouring of fluids [7], or sample scraping [10]. Although IL offers a sample-efficient alternative to RL, its application to chemistry remains under-explored. In chemistry, the difficulty of hand-engineering reward functions for multi-step procedures often makes expert demonstrations a more practical alternative.

Despite the flexibility offered by these learning-based advancements, the widespread adoption of robotic systems in human-centric chemistry laboratories remains contingent on overcoming critical safety hurdles [11], *e.g.*, chemical spillage [12] and reactive collision avoidance in constrained environments [13]. Beyond protecting human scientists from hazardous environments and high-risk procedures, autonomous systems alleviate the burden of repetitive, time-intensive tasks, thereby reallocating human cognitive resources towards complex problem-solving [14]. Addressing the safety challenges within these tasks, from robust robotic manipulation to intelligent decision-making by autonomous agents, is critical for the reliable deployment of these systems [11].

The reliability of IL is fundamentally bounded by the density and diversity of the provided expert demonstrations. A critical challenge in chemistry laboratory-based IL is the distributional shift that occurs when an agent encounters states outside the expert manifold *e.g.*, novel equipment or unexpected reaction dynamics. Because vanilla IL models often fail to distinguish between epistemic uncertainty (lack of data) and aleatoric uncertainty (inherent noise), they are prone to overconfident extrapolation. In a safety-critical environment, as is a chemistry lab, a policy may attempt to execute an action in an out-of-distribution (OOD) state [15], potentially leading to catastrophic failures [11] without any internal mechanism to trigger a safety intervention.

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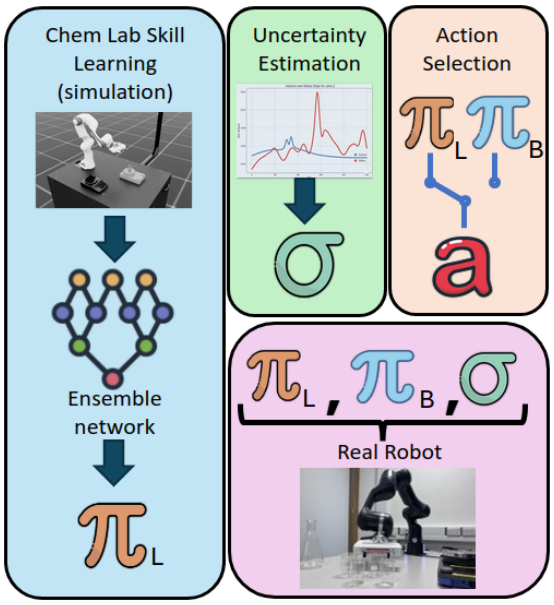


Fig. 1: An overview of our method **SAFE-CHEM**. The framework consists of four stages: (1) training an ensemble-based imitation learning policy (π_L) in simulation for robotic chemistry tasks; (2) estimating the epistemic uncertainty σ through ensemble variance; (3) action selection that monitors σ to gate control between π_L and a fail-safe rule-based backup robot controller (π_B); and (4) real-world demonstration on a robotic system using zero-shot task transfer.

In this work, we propose a modular framework **SAFE-CHEM** that treats failure prevention as a real-time uncertainty-quantification problem, illustrated in Fig. 1. By utilising an ensemble of neural network policies, we monitor epistemic drift during inference to trigger a robust switching mechanism that transitions control back to a fail-safe state whenever the model's confidence falls below a critical threshold, ensuring

that autonomous dexterity does not come at the cost of laboratory safety.

In summary the contributions of this work are:

(1) An ensemble-based imitation learning architecture specifically tailored for chemistry laboratory manipulation, which enhances task robustness by capturing the distribution of expert demonstrations to provide a more resilient policy than traditional single-model approaches;

(2) An uncertainty-aware, policy-agnostic safety framework that monitors variance across ensemble predictions to autonomously identify high-risk states and trigger a switching mechanism, effectively balancing task success with the frequency of autonomous intervention;

(3) An empirical quantification of the relationship between policy confidence and workflow success, providing a systematic analysis of how safety thresholds can be tuned based on the specific risk profile of a laboratory task;

(4) The real-world demonstration through zero-shot transfer of our framework onto a physical robotic manipulator.

II. Related Work

A. Robotics for Laboratory Automation

The adoption of robotic systems in chemistry labs is slowly evolving toward adaptive, modular systems capable of acquiring different, learned laboratory skills. Current approaches for developing these low-level behaviours typically fall into two categories: reinforcement learning (RL) [9], [10] or, more recently, imitation learning. For instance, Xu et al. [16] demonstrated the first application of imitation learning in a materials laboratory by training a a behaviour cloning with a recurrent neural network (BC-RNN) policy on 50 kinaesthetic demonstrations to achieve autonomous cathode painting on fuel cell pellets. More recently, Suzuki et al. [17] proposed TVF-DiT, a compact imitation learning framework that aligns vision and vision-language foundation models with a diffusion transformer-based action expert to enable efficient, high-success-rate robotic laboratory automation on low-VRAM hardware. Despite the promise of these learning-based skills, their deployment in safety-critical chemistry labs is hindered by a lack of failure-prediction mechanisms. Suzuki et al. [17] identified that failures typically occur during the final stages of placement, suggesting a need for OOD detection during high-precision manoeuvres. Unlike these prior works that rely on a single policy, our approach introduces an ensemble-based switching mechanism that monitors action uncertainty across task manifolds to recognise and mitigate potential failures.

B. Uncertainty Estimation and Safety Intervention in Robotic Manipulation

Reliable runtime failure detection is a prerequisite for moving IL policies from controlled experiments to safety-critical laboratories. Recent research has focused on quantifying uncertainty to detect OOD states without requiring explicit failure data. Romer et al. proposed FIPER [18], which uses random network distillation to identify OOD observations. However, such visual-heavy embeddings can be confounded by the optical challenges of real-world chemistry labs, such as transparent glassware and solvents, and occlusions. Similarly, FAIL-Detect [15] employs conformal prediction to calibrate safety bands, but its reliance on temporal consistency makes it sensitive to temporal drift caused by adaptive velocity scaling or a robot approaching its physical limits. Uncertainty quantification has also been explored using ensemble methods. Zhu et al. [19] proposed a heterogeneity uncertainty quantification method for autonomous navigation through crowded intersections, distinguishing between data (aleatoric) and model (epistemic) uncertainties. While effective for 2D lateral and longitudinal control, these methods are primarily designed for low-dimensional action spaces. Our work distinguishes itself from these approaches by treating failure prevention as a pre-emptive uncertainty-quantification problem. While existing systems like FAIL-Detect are often reactive, *i.e.*, identifying a failure only after the state has significantly deviated, our SAFE-CHEM framework monitors epistemic drift to trigger a safety intervention before an irreversible event occurs. Crucially, our method does not require any failure modes to be shown to the robot during training as the failure detection is handled via ensemble variance, which can be continually refined offline as the policy execution improves.

III. Methodology

This work introduces **SAFE-CHEM** an uncertainty-aware robotic skill framework based on deep imitation learning for chemistry laboratories. As illustrated in Fig. 1, our method uses a probabilistic ensemble of deep neural networks to capture the predictive distribution across multiple models trained for the same task, subsequently aggregating their outputs into a single, robust action. This ensemble-based approach nables the estimation of epistemic uncertainty to identify OOD scenarios where the learned policy exhibits low confidence. To maintain operational safety, a threshold-triggered switching mechanism transitions control from the learned policy to a deterministic, rule-based fail-safe system whenever uncertainty exceeds a predefined limit to safeguard task integrity.

A. Imitation Learning for Robotic Manipulation in Chemistry Lab Automation

Imitation learning methods have proven to be effective for training robots to perform complex tasks from expert human demonstrations. We formalise the robot manipulation task as a discrete-time Markov decision processs (MDP), defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma, \rho_0)$, where $\mathcal{S}$ represents the continuous state space, $\mathcal{A}$ represents the continuous action space, $\mathcal{P}(s_{t+1}|s_t, a_t)$ denotes the state transition distribution, $R(s, a, s')$ is the reward function , $\gamma \in [0, 1)$ is the discount factor and ρ_0 is the initial state distribution.

In complex real-world chemistry laboratory environments, the reward function is difficult to analytically specify or sparse in nature. Consequently, we utilise behaviour cloning [20] to learn a policy π_θ directly from a dataset $\mathcal{D} = \{\tau_i\}_{i=1}^N$ consisting of N expert demonstrations. Each demonstration

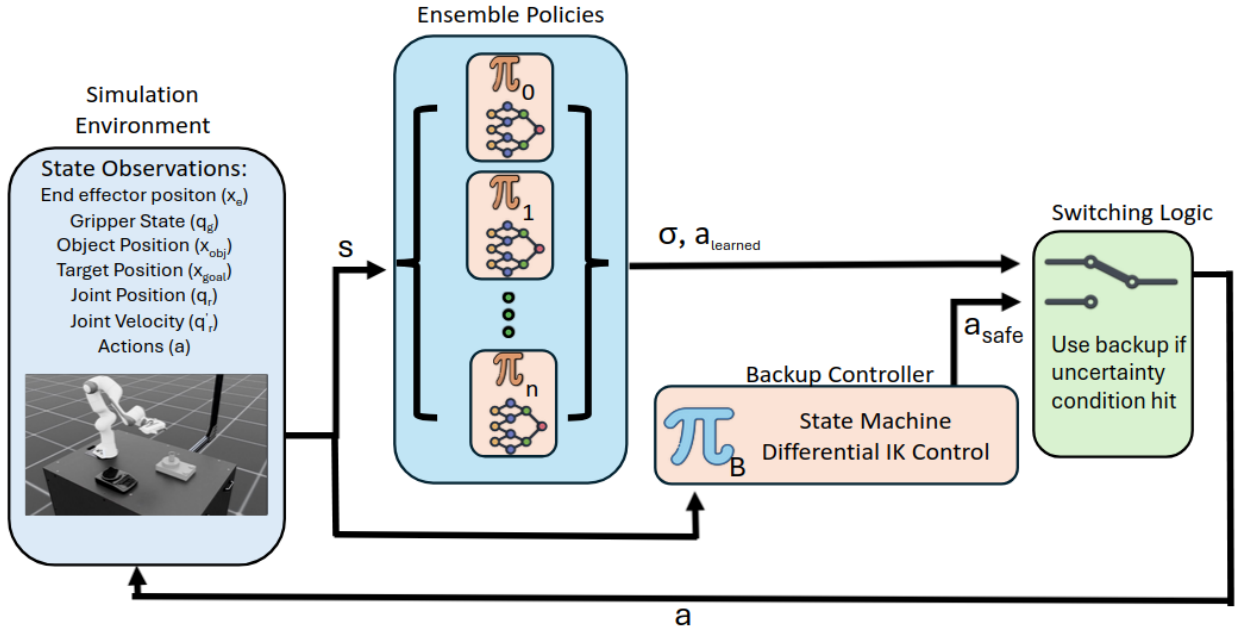


Fig. 2: Overall block diagram

$\tau_i = \{(s_0, a_0), (s_1, a_1), \ldots, (s_T, a_T)\}$ consists of state-action pairs that represent successful task completion.

For continuous action spaces, the objective is to find the optimal policy parameters θ^* by minimising the expected discrepancy between the expert's actions a and the policy's predicted actions $\pi_\theta(s)$. This is formulated as the minimisation of the expected mean squared error, given by Equation 1, where where $\mathbb{E}_{(s,a)\sim\mathcal{D}}$ denotes the expectation over the distribution of state-action pairs (s, a) contained within the expert dataset $\mathcal{D}$, and $\|\cdot\|_2^2$ represents the squared L_2 norm.

$$\theta^* = \arg\min_\theta \mathbb{E}_{(s,a)\sim\mathcal{D}} \left[\|\pi_\theta(s) - a\|_2^2\right] \tag{1}$$

Vanilla BC assumes the Markov property, where the action a_t depends solely on the current state s_t. However, most robot manipulation tasks are inherently partially observable and require historical context to resolve ambiguities. To model these temporal correlations, we employ a recurrent neural network architecture for our policy. Under this framework, the BC-RNN maintains an internal hidden state h_t that serves as a compressed, latent representation of the interaction history. This latent state is updated recursively at each timestep t using a recurrent cell, such that $h_t = f(h_{t-1}, s_t)$. The state h_t is updated at each timestep t using a recurrent cell. The policy π_θ then maps this context-aware representation to the action space, $a_t = \pi_\theta(h_t)$.

B. Uncertainty Quantification via Ensemble Variance

A standard policy remains insufficient for safety-critical chemistry laboratory environments, where model overconfidence in novel or high-risk states could be catastrophic. To mitigate these risks, we employ an ensemble of M neural networks $\{\pi_\theta\}_{m=1}^M$, which quantifies the epistemic uncertainty of the model. Each ensemble member is independently initialised and trained on the expert dataset $\mathcal{D}$, effectively approximating a Bayesian posterior over the policy parameters. For a given latent temporal state h_t, each member m generates an action prediction $a_{t,m} = \pi_{\theta_m}(h_{t,m})$. The final control signal $\bar{a}_t$ is derived from the predictive mean, where $\bar{a}_t = \frac{1}{M}\sum_{m=1}^{M} a_{t,m}$. Our ensemble also captures the temporal divergence of the recurrent latent states through the variance σ_e^2. We hypothesise that the the ensemble's disagreement correlates with the expected prediction error; specifically, as the robot encounters states further from the expert's training manifold, the inter-model variance should increase proportionally with the MSE.

C. Uncertainty-Aware Switching Mechanism

Building upon the uncertainty quantification derived from ensemble variance, we propose a switching mechanism that translates probabilistic signals into discrete control transitions. To distinguish between nominal variance and variance indicative of impending failure, we calibrate component-specific thresholds using a set of reference trajectories. During task execution, we consider the trajectory data for each action component $\mathbf{a}_i \in \mathbb{R}^n$, where $n = 7$ which denotes the degrees-of-freedom (DoF) of the robotic manipulator. For each rollout τ_k, we monitor the ensemble variance $\sigma_{i,t}^2$ at each timestep t for component i using Equation 2.

$$\sigma_{i,t}^2 = \mathbb{E}[(a_{i,t} - \mu_{i,t})^2] \tag{2}$$

In Equation 2, $\mu_{i,t}$ represents the mean action value. To standardise trajectories of variable durations, zero-padding is applied to reach a maximum horizon H. We classify each trajectory τ_k based on task outcome as either successful

($\tau_k \in \mathcal{S}$) or failed ($\tau_k \in \mathcal{F}$). For each successful trajectory, we analyse the variance (σ^2) distributions for each action component to establish component-specific confidence thresholds. We aggregate all non-zero variance values from trajectories in each class to define the reference sets.

$$V_i^{(\mathcal{S})} = \{\sigma_{i,t}^2 \mid \sigma_{i,t}^2 > 0,\, \tau \in \mathcal{S},\, t \in [1, H]\} \tag{3}$$

$$V_i^{(\mathcal{F})} = \{\sigma_{i,t}^2 \mid \sigma_{i,t}^2 > 0,\, \tau \in \mathcal{F},\, t \in [1, H]\} \tag{4}$$

The design of the switching mechanism is predicated on the stochastic divergence of the action ensemble between nominal and failure states. This behaviour is characterised by modelling the density of action variances observed during successful task executions, which defines the boundaries of the agent's competence manifold. By employing Kernel Density Estimation (KDE), we characterise the underlying distribution of these variances to derive a continuous, non-parametric estimate of the success-conditioned density

The Gaussian kernel bandwidth $K_h(u)$ is determined via Scott's Rule [21]:

$$K_h(u) = \frac{1}{\sqrt{2\pi}h} \exp\left(-\frac{u^2}{2h^2}\right) \tag{5}$$

By integrating this density, we obtain the cumulative distribution function (CDF), representing the probability that a successful action exhibits a variance less than or equal to v:

$$F_{\mathcal{S}}(v) = P(\sigma^2 \leq v \mid \tau \in \mathcal{S}) \tag{6}$$

For a specified quantile $q \in (0, 1)$, we define the component-specific confidence threshold $\theta_{i,q}$ as the value below which $q \times 100\%$ of successful variances reside:

$$\theta_{i,q} = \inf\{v : F_{\mathcal{S}}(v) \geq q\} \tag{7}$$

To define the switching logic, we implement a sliding window peak detection algorithm. During online execution, the system monitors the ensemble variance $\sigma_{i,t}^2 = \mathbb{E}[(a_{i,t} - \mu_{i,t})^2]$ for each action component. To prevent false-positive transitions triggered by transient noise or momentary spikes in uncertainty, we implement a sliding window peak detection algorithm. This mechanism ensures that a switch to the safety backup controller occurs only during sustained periods of high variance.

For a given trajectory τ and action component i, the detection signal $D_{i,t}(\tau)$ is defined as positive at timestep t if the number of instances where the variance exceeds the calibrated threshold $\theta_{i,q}$ within a window of size w reaches a limit n_{peaks}:

$$D_{i,t}(\tau) = \left[\sum_{j=t-w}^{t} [\sigma_{i,j}^2 \geq \theta_{i,q}] \geq n_{\text{peaks}}\right] \tag{8}$$

If $D_{i,t}(\tau) = 1$ for any action component $i \in \{1, \ldots, n\}$, the switching logic flags an uncertainty risk. If more than 2 action components return a flag, the system immediately transitions control to the backup policy. The sensitivity of the detection performance depends critically on the hyperparameters w and n_{peaks}, which are empirically tuned to balance safety with task completion.

IV. Experimental Evaluation

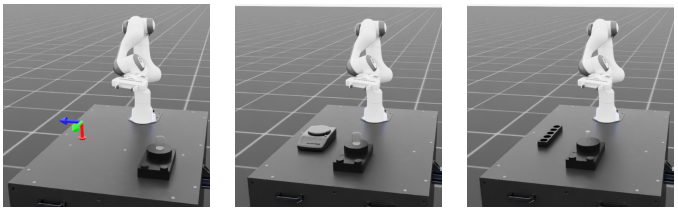
We evaluated our uncertainty-aware imitation learning framework across three chemistry-based manipulation tasks (Fig. 3) to address the following research questions: (1) Does transitioning from a single policy to a distributed ensemble improve task success rate? (2) Can KDE effectively model the stochastic variance of an ensemble to define a robust, quantile-based triggering mechanism for intervention? and (3) How does the framework balance the efficiency of learning-based policies with the reliability of rule-based safety controllers to maximise overall success? Finally, we demonstrate the zero-shot transfer capabilities of our method on a real robotic system.

A. Experimental Setup

All simulation environments were developed within IsaacLab [7], [22]. These were run on a system featuring an AMD Ryzen Threadripper 7970X processor, 128GB of RAM and an NVIDIA GeForce RTX 5090 GPU, operating on Ubuntu 24.04. For consistency across domains, a Franka Research 3 (FR3) robotic manipulator was used in both simulated and real-world configurations. Our learning framework employed an ensemble of Behavioral Cloning paired with a Recurrent Neural Network (BC-RNN) from the RoboMimic framework [23] to effectively capture the temporal dynamics of the manipulation task. The network observes a low-dimensional state vector encompassing the robot's proprioceptive data (eef pose (x_e), joint states(q_r) and velocities ($\dot{q}_r$), gripper state (q_g) and environmental state (object (x_{obj} and target positions (x_{goal})). Temporal dependencies are processed using a 2-layer LSTM with a hidden dimension of 400, trained with a sequence length of 16. The data generation pipeline begins with 10 human demonstrations collected in simulation using a SpaceMouse, which are subsequently expanded by MimicGen [24] to produce a dataset of 4,000 valid trajectories. To account for human demonstration multi-modality, the continuous action space is modeled using a Gaussian Mixture Model (GMM) with 10 modes. To estimate epistemic uncertainty and improve robustness, we utilize a Deep Ensemble approach comprising 15 distinct BC-RNN models. To maximize ensemble diversity, each model is initialized with a unique random seed and trained on a randomly sampled 25% subset of the total demonstration dataset. The models are trained using the Adam optimizer with an initial learning rate of 1e-4 and a batch size of 100 for 1000 epochs.

B. Laboratory Tasks

The experimental evaluation is conducted across three manipulation tasks that represent core operations in an automated chemistry laboratory (Fig. 3). In the **lift task**, the robot needs



(a) **Lift Task** (b) **Pick and place Task** (c) Insertion Task

Fig. 3: The created tasks and setup in the simulated environment and real robot

to pick up an object (*e.g.*, a vial or a beaker) and hold it in a desired position. This task is designed for visual inspection tasks in chemistry experimental workflows [25]. In the **pick and place task**, the robot needs to pick up an object (*e.g.,* a vial or a beaker) from an instrument (*e.g.,* a scale) and place it on another instrument (*e.g.,* IKA RCT plate). The pick and place task is the most predominant manipulation skill in automated chemistry workflows [4], [5]. In the **insertion task**, the robot needs to pick up a vial from an instrument (*e.g.,* IKA RCT plate) and place it inside a rack for transportation or further analysis. This task is a high precision, contact-rich manipulation skill in end-to-end experimental workflows [13].

C. Experiment I: Ensemble Policy Evaluation

To evaluate the performance of the ensembles, we measure the task success rate and the safety incident rate (where the object is dropped or tipped beyond a safe angle). Each task was evaluated for 3×100 episodes and the mean average and standard deviation recorded. This reduces the influence of the random nature of the object spawn location. The task is classified as failed if the object is dropped or tilted beyond $\angle 45$ as this would classify as a safety incident. First we investigate the impact of using an ensemble of policies on the success rate of the tasks.

We evaluate the rollout of the task with an ensemble size of 1 (single policy), 5, 10 and 15 for a maximum of 5000 steps. Table I shows the results of the study into ensemble size on task success rate.

TABLE I: Success rate (%) across 3×100 rollouts for varying ensemble sizes in the lift, place, and insertion tasks.

Size	**Lift** ($\% \pm$ SD)	**Place** ($\% \pm$ SD)	**Insert** ($\% \pm$ SD)
1	75.7 ± 2.5	67.7 ± 9.1	21.3 ± 0.6
5	88.3 ± 4.2	63.7 ± 7.6	51.7 ± 4.0
10	$\mathbf{93.0 \pm 1.7}$	69.0 ± 8.5	$\mathbf{55.0 \pm 9.8}$
15	92.0 ± 1.0	$\mathbf{83.0 \pm 2.7}$	42.0 ± 4.0

Table I shows a significant increase in the success rate of the task execution using an ensemble of policies for the all tasks investigated. During the lift and place tasks, we observe that the period of the task with the highest uncertainty is the grasping step, however in the insert task the uncertainty is greatest towards the end of the task execution due to the high level of manipulation precision required. We did not observe a consistent improvement in task completion with increasing ensemble size, however the optimal ensemble size for all the tasks investigated were at least 10, suggesting that for most tasks and ensemble size of 10 or greater will lead to optimal task completion.

In the very simple lift task, the increasing the number of policies collaborating does not provide significant benefit above an ensemble size of 10, whereas for the pick and place task, the impact of the increased ensemble size is not significant until we reach 15 policies. This may be due to an outlying place policy that is not effectively buffered until the ensemble size reaches 15. For the insertion task, the addition of policies above an ensemble size of 5 did not show an improvement in the task success rate and above an ensemble of 10, the task performance decreased. This may be due to the larger ensemble of polices predicting a more conservative action to be taken, tending to cause smaller action to be taken and the task to timeout.

D. Experiment II: Calibrating the Switching Logic

We evaluate the proposed switching logic on a 7-DoF robotic manipulator. We collect trajectory data for the ensemble policy rollout with no intervention and record the task execution as successful or failed. During task execution, we collect trajectory data for each action component $\mathbf{a}_i \in \mathbb{R}^7$ representing the 7-DOF robotic manipulator. As established in Section X, we applied KDE to the aggregated variance data. Empirical observations (Fig. 4) show a distinct shift toward lower variances for successful trajectories ($\mathcal{S}$) compared to failed ones ($\mathcal{F}$), validating the assumption that the ensemble policy maintains higher consistency during confident, successful actions. We investigated three confidence thresholds at specified quantiles q for a conservative (0.90), moderate (0.95) and relaxed (0.99) uncertainty limit.

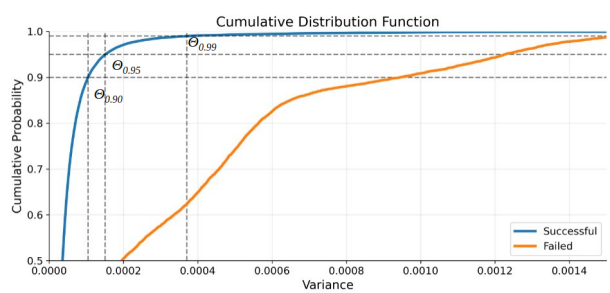


Fig. 4: Confidence thresholds for the $\theta_{0.90}$, $\theta_{0.95}$ and $\theta_{0.99}$ bounds

Due to certain actions such as grasping being inherently uncertain, we trigger a switch only when the condition in Eq. 8 is met for more than two action components. We utilised a sliding window size $w \in [8, 10]$ and performed a grid search over $n_{\text{peaks}} \in \{1, \ldots, 10\}$.

Detection performance was measured via Detection Rate (True Positives), False Alarm Rate (False Positives), and F1-

Score. As shown in Table II, an n_{peaks} value of 8 consistently provided the highest F1-score across "Lift", "Place", and "Insert" tasks when using the conservative $\theta_{0.90}$ limit. This configuration was selected for final implementation to maximise failure detection while minimising unnecessary interruptions.

E. Experiment III: Confidence Level Intervention

Next we investigate the success rate at different intervention levels (θ_q) for the different tasks in simulation. First we find the most conservative confidence level of $\theta_{0.90}$ of the successful task uncertainties and the window that best distinguishes between the successful and failed task executions for each task. As part of the hybrid control method, we also create a state machine based differential IK controller for each of the tasks to act as the safety backup controller. This task controller is slower to complete the task then the learning based policy but is not subject to uncertainty in the action (as the actions will always benefit task completion).

The performance of the hybrid uncertainty-aware control for the simulation task is shown in table III, the intervention rate of the switching logic is shown in table IV and the critical failure rate is shown in table V.

1) Task success rate: For all tasks, intervention when the uncertainty switching logic is triggered resulted in improved task success rate, with all executions of the task with the hybrid controller out performing the ensemble with no intervention. For the lift task, the best performing model was with an ensemble of 10 policies and the most relaxed confidence level ($\theta_{0.99}$) with an average success rate of 99.3% across the 3 test rollouts. The ensembles with 10 or more polices have a high base success rate for the lift task (the ensemble of polices alone has a high success rate) and so the improvement seen by using the hybrid uncertainty aware controller is reduced when compared to the smaller ensemble and single policies.

The pick and place task also shows a significant improvement in the task success when using the uncertainty aware intervention, with all tests with intervention outperforming the ensemble alone. The best performing run was with the mid-level confidence interval and an ensemble size of 5 policies in our experiment. However, the performance of the ensemble size of 10 is within the standard deviation. As the pick and place task is harder for the agent to execute, the success rate of the policy alone is lower than with the simple lift task. For the most challenging task - the vial insertion task, we observe that the best task success was with the most conservative uncertainty limit ($\theta_{0.90}$) for all ensemble sizes. As this task is significantly more difficult for the robot to complete due to the high level of precision required, the lower confidence level causes the backup controller to be activated more frequently, leading to a higher success rate. We observed during this task that the learned policy would sometimes create unrecoverable states that were not critical safety incidents. These occurred when the vial was slightly misplaced in the vial rack meaning that the vial was in the vial rack but not completely inserted. The backup controller was not able to reach the vial to grip it and place correctly in the vial rack- leading to the attempt being failed but no safety incident occurred.

2) Switching Intervention Rate: As expected, the intervention rate of the switching logic reduces with the increasing uncertainty limit for most of the tasks. At the most conservative uncertainty limit $\theta_{0.90}$, the larger ensemble has a higher intervention rate, however as this limit relaxes, the intervention rate of the switching logic for the larger ensemble reduces to below that of the smaller policies. For the challenging vial insertion task, the intervention rate between the $\theta_{0.90}$ and $\theta_{0.95}$ was not significantly different for the ensembles of fewer than 15 polices. This suggests that for difficult tasks, the larger ensemble produced a better representation of the uncertainty of the task actions.

3) Safety Failure Rate: For the simple lift task, increasing the number of policies in the ensemble did not significantly benefit the rate of safety violations. This was due to, during task execution, it was observed that the ensemble policies alone tended to fail due to timeout as the robot did not attempt to pick up the object. With the hybrid controller, the backup controller was triggered to intervene at an inconvenient time, resulting in the object getting tipped and causing a safety violation. We observe that for all of the tasks, having the ensemble of policies tends to reduce the safety failure rate, even without the hybrid control. This is largely due to the ensemble of policies tending to predict a more conservative action than a single policy alone. With lift task, the safety violation rate is significantly reduced by using the ensemble of policies, the pattern of the reduction in safety violation rate follows the success rate in this task- suggesting that more complex tasks are more suited to using a larger ensemble of policies with a more conservative intervention criteria.

F. Real-World Deployment

To validate the physical transferability and effectiveness of our learned control policies, we conducted qualitative real-world experiments using a physical Franka FP3 robotic arm. We evaluated the system on the pick-and-place task to test the policy's ability to approach, grasp, and reliably manipulate objects outside of simulation. The control architecture leveraged a zero-shot sim-to-real configuration, where inference was performed using the ensemble policy deployed within the Isaac Lab environment. At each control step, the inferred joint position targets were transmitted via a low-latency UDP connection to a custom ROS 2 bridge node running on the robot's control workstation.

This bridge processed the incoming data and published the simulated joint targets as ROS2 messages to a joint position controller, while parallel gripper commands were dispatched to the gripper action server. We utilized MoveIt2 to interface with the low-level joint tracking controllers, providing a safe and robust execution pipeline for the physical hardware. During qualitative testing, the physical robot successfully completed the full pick-and-place task in the real world. This successful execution demonstrates that the behaviour cloning policies and the integration of our uncertainty-aware backup controllers,

TABLE II: Performance of the uncertainty peak detection algorithm across different window sizes w and peak counts n_{peaks}.

	True Positives			**False Positives**			**F1-Score**		
n_{peaks}	Lift w=10	Place w=10	Insert w=9	Lift w=10	Place w=10	Insert w=9	Lift w=10	Place w=10	Insert w=9
1	1.00	1.00	1.00	1.00	1.00	1.00	0.67	0.67	0.67
2	1.00	1.00	1.00	1.00	1.00	1.00	0.67	0.67	0.67
3	1.00	1.00	1.00	0.97	1.00	1.00	0.67	0.67	0.67
4	1.00	1.00	0.99	0.82	1.00	0.86	0.71	0.67	0.69
5	1.00	0.94	0.97	0.63	0.75	0.67	0.76	0.70	0.74
6	1.00	0.88	0.95	0.50	0.39	0.48	0.80	0.78	0.78
7	1.00	0.88	0.93	0.37	0.17	0.36	0.84	0.86	0.82
8	1.00	0.88	0.91	0.23	0.05	0.20	**0.90**	**0.91**	**0.87**
9	0.88	0.84	0.87	0.12	0.02	0.13	0.88	0.90	0.87
10	0.75	0.80	0.81	0.03	0.01	0.09	0.84	0.89	0.85

TABLE III: Task success rate (%) across varying ensemble sizes and confidence thresholds θ_q. "None" indicates the baseline performance without the switching mechanism.

	Success rate (Lift)				**Success rate (Place)**				**Success rate (Insert)**			
Size	None	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$	None	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$	None	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$
1	75.7 ± 2.5	–	–	–	67.7 ± 9.0	–	–	–	21.3 ± 0.6	–	–	–
5	83.3 ± 4.2	96.0 ± 2.0	97.7 ± 2.1	96.3 ± 1.5	63.7 ± 7.6	94.3 ± 1.5	$\mathbf{98.3 \pm 1.2}$	89.3 ± 1.5	51.7 ± 4.0	60.0 ± 4.4	58.3 ± 3.8	59.0 ± 2.7
10	93.0 ± 1.7	96.3 ± 0.6	99.0 ± 1.0	$\mathbf{99.3 \pm 0.6}$	69.0 ± 8.5	96.3 ± 0.6	97.0 ± 2.0	86.4 ± 4.0	55.0 ± 9.8	67.3 ± 3.5	62.7 ± 5.5	58.0 ± 6.1
15	92.0 ± 1.0	98.0 ± 1.0	99.0 ± 0.0	98.7 ± 1.5	83.0 ± 2.7	96.7 ± 1.5	96.0 ± 1.7	91.0 ± 1.7	42.0 ± 4.0	$\mathbf{68.0 \pm 1.7}$	64.3 ± 4.2	59.7 ± 2.5

TABLE IV: Switching logic intervention rate (%) for different tasks and allowed confidence thresholds θ_q.

	Intervention rate (Lift)			**Intervention rate (Place)**			**Intervention rate (Insert)**		
Size	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$
5	24.3 ± 5.1	16.0 ± 4.6	9.0 ± 3.6	43.3 ± 11.9	30.0 ± 7.9	16.1 ± 5.1	28.0 ± 6.6	30.0 ± 2.6	23.0 ± 7.2
10	27.7 ± 1.5	10.3 ± 1.2	10.7 ± 4.6	47.0 ± 6.9	26.3 ± 1.2	15.3 ± 2.1	31.3 ± 4.0	32.0 ± 5.3	21.3 ± 1.5
15	30.7 ± 2.5	11.3 ± 1.5	6.7 ± 4.1	28.0 ± 4.6	10.3 ± 2.5	8.0 ± 0.0	52.0 ± 2.0	41.0 ± 5.3	31.3 ± 2.5

TABLE V: Rate of critical safety violations (%) (e.g., dropped objects or tilts $> 45°$) across varying ensemble sizes and confidence thresholds θ_q.

	Safety failure rate (Lift)				**Safety failure rate (Place)**				**Safety failure rate (Insert)**			
Size	None	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$	None	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$	None	$\theta_{0.90}$	$\theta_{0.95}$	$\theta_{0.99}$
1	3.3 ± 0.6	–	–	–	11.3 ± 5.0	–	–	–	62.3 ± 5.1	–	–	–
5	1.7 ± 2.9	1.0 ± 1.0	0.3 ± 0.6	3.7 ± 1.5	8.0 ± 1.0	4.0 ± 2.0	1.7 ± 1.2	2.0 ± 1.7	25.3 ± 1.2	33.0 ± 3.6	34.7 ± 1.5	33.3 ± 1.2
10	0.0 ± 0.0	0.0 ± 0.0	0.7 ± 0.6	0.7 ± 0.6	6.0 ± 3.0	1.7 ± 0.6	3.0 ± 2.0	3.0 ± 1.0	21.0 ± 7.0	24.7 ± 2.3	26.7 ± 4.0	29.3 ± 3.1
15	2.3 ± 1.5	1.3 ± 0.6	1.0 ± 0.0	1.3 ± 1.5	3.0 ± 1.0	2.0 ± 1.0	2.7 ± 1.2	2.0 ± 1.0	23.3 ± 0.6	22.7 ± 4.6	27.7 ± 4.9	25.3 ± 0.6

developed entirely within the simulated environment, can be robustly deployed to physical hardware without requiring any supplementary real-world fine-tuning.

V. Conclusion

References

[1] A. I. Cooper, P. Courtney, K. Darvish, M. Eckhoff, H. Fakhruldeen, A. Gabrielli, A. Garg, S. Haddadin, K. Harada, J. Hein, M. Hübner, D. Knobbe, G. Pizzuto, F. Shkurti, R. Shrestha, K. Thurow, R. Vescovi, B. Vogel-Heuser, Ádám Wolf, N. Yoshikawa, Y. Zeng, Z. Zhou, and H. Zwirnmann, "Accelerating discovery in natural science laboratories with ai and robotics: Perspectives and challenges," *Science Robotics*, vol. 10, no. 106, p. eadv7932, 2025. [Online]. Available: https://www.science.org/doi/abs/10.1126/scirobotics.adv7932

[2] N. Yoshikawa, Y. Asano, D. N. Futaba, K. Harada, T. Hitosugi, G. N. Kanda, S. Matsuda, Y. Nagata, K. Nagato, M. Naito, T. Natsume, K. Nishio, K. Ono, H. Ozaki, W. Shin, J. Shiomi, K. Shizume, K. Takahashi, S. Takeda, I. Takeuchi, R. Tamura, K. Tsuda, and Y. Ushiku, "Self-driving laboratories in japan," *Digital Discovery*, vol. 4, pp. 1384–1403, 2025. [Online]. Available: http://dx.doi.org/10.1039/D4DD00387J

[3] J. Li, C. Ding, D. Liu, L. Chen, and J. Jiang, "Autonomous laboratories in china: an embodied intelligence-driven platform to accelerate chemical discovery," *Digital Discovery*, vol. 4, pp. 1672–1684, 2025. [Online]. Available: http://dx.doi.org/10.1039/D5DD00072F

[4] B. Burger, P. M. Maffettone, V. V. Gusev, C. M. Aitchison, Y. Bai, X. Wang, X. Li, B. M. Alston, B. Li, R. Clowes, N. Rankin, B. Harris, R. S. Sprick, and A. I. Cooper, "A mobile robotic chemist," *Nature*, vol. 583, no. 7815, pp. 237–241, Jul 2020. [Online]. Available: https://doi.org/10.1038/s41586-020-2442-2

[5] A. M. Lunt, H. Fakhruldeen, G. Pizzuto, L. Longley, A. White, N. Rankin, R. Clowes, B. Alston, L. Gigli, G. M. Day, A. I. Cooper, and S. Y. Chong, "Modular, multi-robot integration of laboratories: an autonomous workflow for solid-state chemistry," *Chem. Sci.*, vol. 15, pp. 2456–2463, 2024. [Online]. Available: http://dx.doi.org/10.1039/D3SC06206F

[6] Y. Xie, K. Sattari, C. Zhang, and J. Lin, "Toward autonomous laboratories: Convergence of artificial intelligence and experimental automation," *Progress in Materials Science*, vol. 132, no. 0079-6425, p. 101043, 2023.

[7] K. Darvish, A. Sohal, A. Mandal, H. Fakhruldeen, N. Radulov, Z. Zhou, S. Veeramani, J. Choi, S. Han, B. Zhang, J. Chae, A. Wright, Y. Wang,

H. Darvish, Y. Zhao, G. Tom, H. Hao, M. Bogdanovic, G. Pizzuto, A. I. Cooper, A. A. Guzik, F. Shkurti, and A. Garg, "Matterix: toward a digital twin for robotics-assisted chemistry laboratory automation," *Nature Computational Science*, vol. 6, pp. 67 – 82, 2025. [Online]. Available: https://api.semanticscholar.org/CorpusID:284363658

[8] Y. Kadokawa, M. Hamaya, and K. Tanaka, "Learning robotic powder weighing from simulation for laboratory automation," in *The IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.

[9] N. Radulov, A. Wright, T. Little, A. I. Cooper, and G. Pizzuto, "Flip: Flowability-informed powder weighing," 2025. [Online]. Available: https://arxiv.org/abs/2506.03896

[10] G. Pizzuto, H. Wang, H. Fakhruldeen, B. Peng, K. S. Luck, and A. I. Cooper, "Accelerating laboratory automation through robot skill learning for sample scraping*," in *2024 IEEE 20th International Conference on Automation Science and Engineering (CASE)*, 2024, pp. 2103–2110.

[11] S. X. Leong, C. Griesbach, R. Zhang, K. Darvish, Y. Zhao, A. Mandal, Y. Zou, H. Hao, V. Bernales, and A. Aspuru-Guzik, "Steering towards safe self-driving laboratories," 12 2024.

[12] S. Veeramani, Z. Zhou, F. Munguia-Galeano, H. Fakhruldeen, T. Roddelkopf, M. F. R. Al-Okby, K. Thurow, and A. I. Cooper, "Prevent: Proactive risk evaluation and vigilant execution of tasks for mobile robotic chemists using multi-modal behavior trees," 2025. [Online]. Available: https://arxiv.org/abs/2510.21438

[13] L. Longley, F. Munguia-Galeano, Y. Han, R. Clowes, S. Vijayakrishnan, A. Edwards, G. Pizzuto, H. Fakhruldeen, and A. Cooper, "Robinhood: A robotic chemist in a fume hood," *ChemRxiv*, vol. 2026, no. 0112, 2026. [Online]. Available: https://chemrxiv.org/doi/abs/10.26434/chemrxiv-2026-s2619

[14] Y. Xie, K. Sattari, C. Zhang, and J. Lin, "Toward autonomous laboratories: Convergence of artificial intelligence and experimental automation," *Progress in Materials Science*, vol. 132, p. 101043, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0079642522001244

[15] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. Ambrus, H. Nishimura, and M. Itkina, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," in *Proceedings of Robotics: Science and Systems (RSS)*, 2025.

[16] X. Xu, Y. Gu, T. Zhang, J. Yu, and S. Skinner, "Revolutionizing clean energy labs: Robotic imitation learning for efficient fabrication ai-powered electrical units assembly platform," *Energy and AI*, vol. 21, p. 100517, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666546825000497

[17] K. Suzuki, H. Nakamurama, K. Miyamoto, and T. Ogata, "Compact task-aligned imitation learning for laboratory automation," 2026. [Online]. Available: https://arxiv.org/abs/2603.01110

[18] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, "Failure prediction at runtime for generative robot policies," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[19] Z. Zhu, S. Wang, and H. Zhao, "Uncertainty-aware deep imitation learning and deployment for autonomous navigation through crowded intersections," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 13 980–13 987.

[20] D. Pomerleau, "Alvinn: An autonomous land vehicle in a neural network," in *Proceedings of (NeurIPS) Neural Information Processing Systems*, D. Touretzky, Ed. Morgan Kaufmann, December 1989, pp. 305 – 313.

[21] B. W. Silverman, *Density estimation for statistics and data analysis*. Routledge, 2018.

[22] NVIDIA, :, M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrügg, N. Rudin, L. Wawrzyniak, M. Rakhsha, A. Denzler, E. Heiden, A. Borovicka, O. Ahmed, I. Akinola, A. Anwar, M. T. Carlson, J. Y. Feng, A. Garg, R. Gasoto, L. Gulich, Y. Guo, M. Gussert, A. Hansen, M. Kulkarni, C. Li, W. Liu, V. Makoviychuk, G. Malczyk, H. Mazhar, M. Moghani, A. Murali, M. Noseworthy, A. Poddubny, N. Ratliff, W. Rehberg, C. Schwarke, R. Singh, J. L. Smith, B. Tang, R. Thaker, M. Trepte, K. V. Wyk, F. Yu, A. Millane, V. Ramasamy, R. Steiner, S. Subramanian, C. Volk, C. Chen, N. Jawale, A. V. Kuruttukulam, M. A. Lin, A. Mandlekar, K. Patzwaldt, J. Welsh, H. Zhao, F. Anes, J.-F. Lafleche, N. Moënne-Loccoz, S. Park, R. Stepinski, D. V. Gelder, C. Amevor, J. Carius, J. Chang, A. H. Chen, P. de Heras Ciechomski, G. Daviet, M. Mohajerani, J. von Muralt, V. Reutskyy, M. Sauter, S. Schirm, E. L. Shi, P. Terdiman, K. Vilella, T. Widmer, G. Yeoman, T. Chen, S. Grizan, C. Li, L. Li, C. Smith, R. Wiltz, K. Alexis, Y. Chang, D. Chu, L. J. Fan, F. Farshidian, A. Handa, S. Huang, M. Hutter, Y. Narang, S. Pouya, S. Sheng, Y. Zhu, M. Macklin, A. Moravanszky, P. Reist, Y. Guo, D. Hoeller, and G. State, "Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning," 2025. [Online]. Available: https://arxiv.org/abs/2511.04831

[23] A. Mandlekar, D. Xu, J. Wong, S. Nasiriany, C. Wang, R. Kulkarni, L. Fei-Fei, S. Savarese, Y. Zhu, and R. Martín-Martín, "What matters in learning from offline human demonstrations for robot manipulation," in *Conference on Robot Learning (CoRL)*, 2021.

[24] A. Mandlekar, S. Nasiriany, B. Wen, I. Akinola, Y. Narang, L. Fan, Y. Zhu, and D. Fox, "Mimicgen: A data generation system for scalable robot learning using human demonstrations," in *7th Annual Conference on Robot Learning*, 2023.

[25] M. Walker, G. Pizzuto, H. Fakhruldeen, and A. I. Cooper, "Go with the flow: deep learning methods for autonomous viscosity estimations," *Digital Discovery*, vol. 2, pp. 1540–1547, 2023. [Online]. Available: http://dx.doi.org/10.1039/D3DD00109A